

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA
S2 B.Tech CSE-AI
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 29

Team Members :

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1. Problem Title : Online Donation Management System

2. Problem Description : Manual donation tracking reduces transparency. Donors make contributions online. Transactions are recorded securely. Donation usage is tracked transparently. Reports are generated for accountability. This builds donor trust.

3. Identified Entities (Classes)

Class Name	Description
Donor	Person who makes donations
Organization	Receives and uses donations
Admin	Manages and monitors the system
Donation	Handles donation transaction
Report	Generates donation and usage reports
Payment	Handles payment

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Donor	donorID, name, email, phone, address	register(), login(), makeDonation(), viewHistory()
Organization	orgID, orgName, description, totalFunds	receiveDonation(), updateUsage(), viewFunds()
Admin	adminID, username, password	monitorDonations(), generateReports(), manageUsers()
Donation	donationID, donorID, amount, date, paymentMethod	recordDonation(), validateDonation()
Report	reportID, date, totalAmount	generateReport(), exportReport()
Payment	paymentID, amount, paymentType, status	processPayment(), verifyPayment()